

SUPPLEMENTARY INFORMATION (SI)

Table 1 Mean (standard deviation in parentheses) and range in body mass (in mg), wing length (in mm) and proboscis length (in mm) in the pooled dataset of butterflies and within the five families.

		Body mass	Wing length	Proboscis length
Pooled (n=251)	Mean	147.9 (158.1)	28.7 (13.05)	12.8 (7.4)
	Range	3.3 - 1459.3	8.8 - 71.1	3.1 - 41
Papilionidae (n=26)	Mean	385.2 (252.51)	52.1 (11.65)	21.2 (8.35)
	Range	85.2 - 1459.3	20.1 - 78.1	9.29 - 35.7
Nymphalidae (n=100)	Mean	174.2 (139.13)	32.7 (8.62)	11.2 (3.36)
	Range	13.9 - 777.5	17.1 - 53.21	5 - 20.3
Pieridae (n=30)	Mean	110.5 (88.64)	30.8 (8.13)	13.9 (5.92)
	Range	9.5 - 367.2	15.02 - 48.9	8.25 - 35.68
Hesperiidae (n=45)	Mean	91.1 (61.29)	18 (4.30)	18 (9.57)
	Range	12.4 - 378.5	10.5 - 31.99	8 - 41
Lycaenidae (n=50)	Mean	45.5 (45.32)	16.8 (4.35)	6.1 (1.40)
	Range	3.3 - 189.6	8.8 - 30.5	3.1 - 9.9

Table 2 Results from linear models with logged wing length (in mm) and logged proboscis length (in mm) as the response variables and logged body mass (in mg), family, habitat and interaction between body mass and family as predictors.

	Estimate mean $\pm$ SE	<i>t</i> -statistic	<i>p</i>
Wing Length			
Intercept (Hesperiidae-Ecotone)	1.47 $\pm$ 0.117	12.58	<0.001
log (Body mass)	0.33 $\pm$ 0.027	12.53	<0.001
Lycaenidae	0.55 $\pm$ 0.133	4.16	<0.001
Nymphalidae	0.70 $\pm$ 0.133	5.26	<0.001
Papilionidae	-0.60 $\pm$ 0.306	-1.97	0.051
Pieridae	0.71 $\pm$ 0.166	4.26	<0.001
Habitat Forest	-0.05 $\pm$ 0.017	-2.95	0.004
Habitat Open	-0.07 $\pm$ 0.025	-2.92	0.004
log (Body mass):Lycaenidae	-0.09 $\pm$ 0.032	-2.80	0.006
log (Body.mass):Nymphalidae	-0.06 $\pm$ 0.030	-1.97	0.05
log (Body.mass):Papilionidae	0.20 $\pm$ 0.056	3.64	<0.001
log (Body.mass):Pieridae	-0.04 $\pm$ 0.037	-1.20	0.232
Proboscis length			
Intercept (Hesperiidae-Ecotone)	0.47 $\pm$ 0.255	1.84	0.067
log (Body mass)	0.53 $\pm$ 0.058	9.07	<0.001
Lycaenidae	0.87 $\pm$ 0.289	3.02	0.003
Nymphalidae	0.25 $\pm$ 0.293	0.85	0.396
Papilionidae	-1.94 $\pm$ 0.668	-2.90	0.004
Pieridae	0.88 $\pm$ 0.363	2.42	0.016
Habitat Forest	-0.002 $\pm$ 0.037	-0.08	0.939
Habitat Open	0.08 $\pm$ 0.055	1.52	0.129
log (Body mass):Lycaenidae	-0.40 $\pm$ 0.071	-5.62	<0.001
log (Body.mass):Nymphalidae	-0.19 $\pm$ 0.065	-2.89	0.004
log (Body.mass):Papilionidae	0.24 $\pm$ 0.122	2	0.047
log (Body.mass):Pieridae	-0.26 $\pm$ 0.081	-3.20	0.002